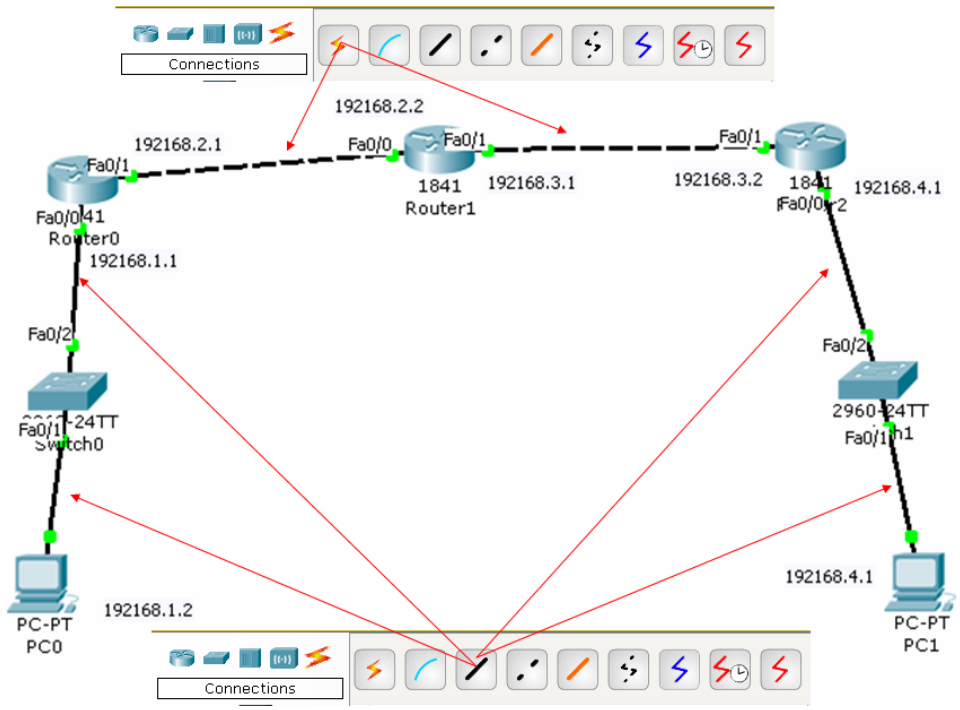
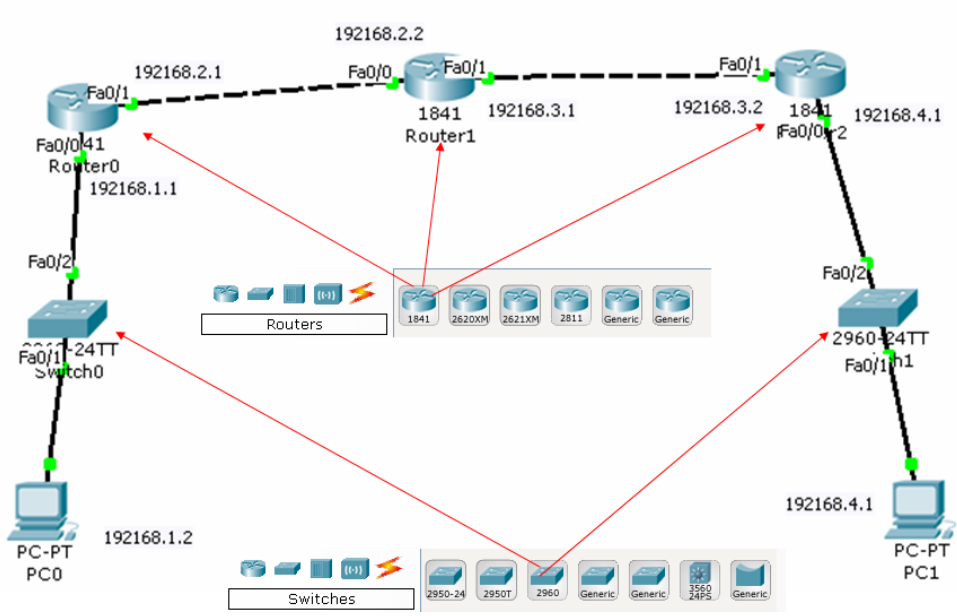
Expt-5
Routing through Multiple Routers in Series

This experiment deals with routing packets through three routers in series.

Step-1
Implement the circuit shown below where the steps are visualized from the figures.





Step-2

Configure the Router-0 like:

Continue with configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

Step-3

Configure the Router-1 like:

Router>enable

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.2.2 255.255.255.0
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

Step-4
Configure the Router-2 like:

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#ip address 192.168.3.2 255.255.255.0
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.4.1 255.255.255.0
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
```

Step-4
Configure the Router-0 again like:

```
Router(config)#ip route 192.168.3.0 255.255.255.0 192.168.2.2
Router(config)#ip route 192.168.4.0 255.255.255.0 192.168.2.2
```

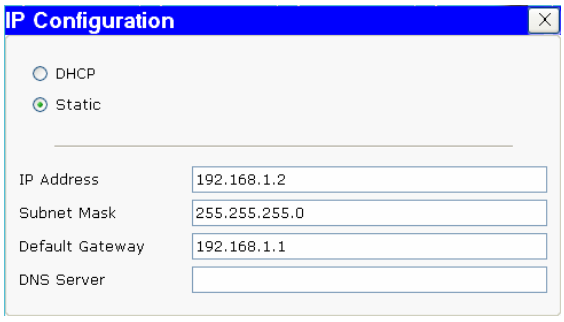
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

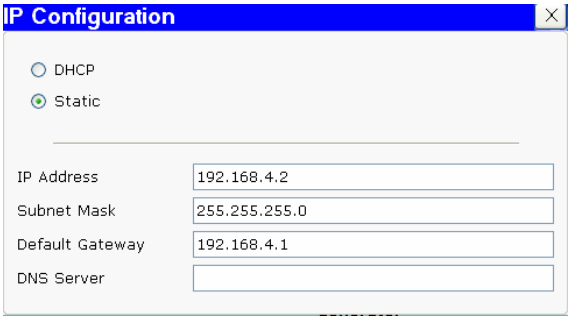
Step-5
Configure the Router-1 again like:
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.2.1
Router(config)#ip route 192.168.4.0 255.255.255.0 192.168.3.2
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

Step-6
Configure the Router-2 again like:

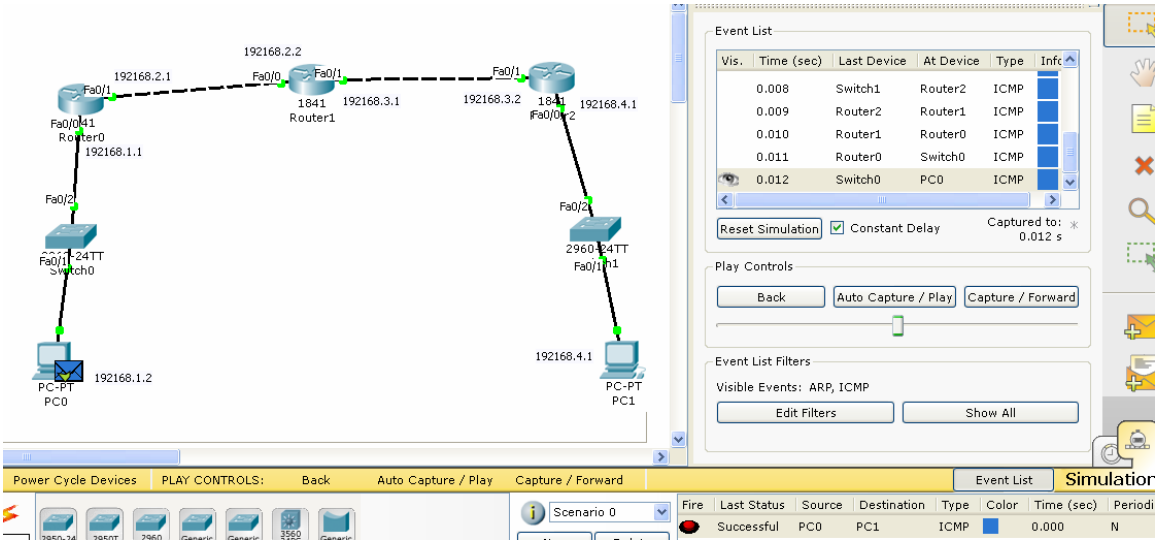
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.3.1
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.3.1
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

Step-7
Configure PC-0 and PC-1 like:





Step-8
Verify the network sending ICMP packet under simulation mode.


192168.2.2
192168.2.1
Fa0/0
Fa0/1
Fa0/1
Fa0/1
1841
Router1
192168.3.1
192168.3.2
1841
192168.4.1
Fa0/0
Router0
192168.1.1
Fa0/2
Fa0/2
Fa0/1
Switch0
Fa0/1
192168.4.1
192168.1.2
PC-PT
PC0
PC-PT
PC1
Event List
Vis.
Time (sec)
Last Device
At Device
Type
0.008
Switch1
Router2
ICMP
0.009
Router2
Router1
ICMP
0.010
Router1
Router0
ICMP
0.011
Router0
Switch0
ICMP
0.012
Switch0
PC0
ICMP
Reset Simulation
Constant Delay
Captured to:
0.012 s
Play Controls
Back
Auto Capture / Play
Capture / Forward
Event List Filters
Visible Events: ARP, ICMP
Edit Filters
Show All
Power Cycle Devices
PLAY CONTROLS:
Back
Auto Capture / Play
Capture / Forward
Event List
Simulation
Scenario 0
Fire
Last Status
Source
Destination
Type
Color
Time (sec)
Periodi
Successful
PC0
PC1
ICMP
0.000
N